O11e - Optical Polarisation

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1 Introduction

Polarization describes the temporal and spatial behaviour of the electric-field vector of a transverse electromagnetic wave. Depending on the relative amplitudes and phase difference of two orthogonal electric-field components, light can be linearly, circularly, or elliptically polarized. In addition, natural light may be unpolarized or may consist of a mixture of polarized and unpolarized components. The analysis of polarization is therefore an important method for characterizing optical systems and materials.

A linear analyzer can be used to investigate the linear components of an incident light beam by measuring the transmitted intensity as a function of the analyzer angle. For linearly polarized light, the transmitted intensity follows Malus' law and varies periodically with the analyzer orientation. Circularly polarized and unpolarized light, however, both produce an analyzer-angle-independent intensity. A quarter-wave plate is therefore required to distinguish these states. It introduces a phase difference of approximately $\pi/2$ between two orthogonal electric-field components and can consequently transform linear polarization into elliptical or circular polarization, and circular or suitable elliptical polarization into linear polarization.

The aim of this experiment was first to investigate the effect of a quarter-wave plate on initially linearly polarized light. A yellow LED with a wavelength of approximately

$$\lambda = 590\,\mathrm{nm}$$

was used as the light source. The transmitted light was measured with a photodiode while the linear analyzer was rotated through a complete angular interval. Measurements were performed for quarter-wave-plate orientations of

$$\phi = 0^\circ, \qquad 30^\circ, \qquad 45^\circ,$$

and the resulting angular intensity dependences were compared with the predictions obtained from Jones calculus.

In the second part of the experiment, three unknown optical elements, referred to as black boxes, were characterized. For each black box, complete analyzer scans were recorded both without and, where necessary, with the quarter-wave plate. The observed modulation, the position of the intensity extrema, and whether complete extinction could be achieved were used to determine the corresponding polarization state. Particular attention was given to distinguishing fully polarized states from mixtures containing an unpolarized component.

The measured photodiode voltage was taken to be proportional to the incident optical intensity. The experimental results were evaluated using theoretical polarization models, sinusoidal fits, physical consistency checks, and an analysis of both random and systematic uncertainties. Important uncertainty sources included the manual adjustment of the analyzer and quarter-wave plate, fluctuations of the measured voltage, the detector background signal, optical absorption by the quarter-wave plate, and deviations of the real optical components from ideal behaviour.

2 Theoretical Analysis

2.1 Overview and angle conventions

For a monochromatic electromagnetic wave propagating in the z-direction, the electric field lies in the transverse xy-plane. Its polarization state can be described by the Jones vector

$$\mathbf{J} = \begin{pmatrix} J_x \\ J_y \end{pmatrix}. \tag{1}$$

In this report, all angles are measured relative to the vertical y-axis. The angle of the incident linear polarization is denoted by α, the fast-axis angle of the quarter-wave plate by ϕ, and the transmission-axis angle of the analyzer by θ.

A linearly polarized wave at an angle α is represented by

$$\mathbf{J}_\alpha = \begin{pmatrix} \sin\alpha \\ \cos\alpha \end{pmatrix}. \tag{2}$$

Right- and left-circularly polarized light are represented, using the convention of the experiment manual, by

$$\mathbf{J}_+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad \mathbf{J}_- = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}. \tag{3}$$

For the LED wavelength of $\lambda = 590\,\mathrm{nm}$, the optical frequency is approximately

$$f = \frac{c}{\lambda} \approx 5.1 \times 10^{14}\,\mathrm{Hz}. \tag{4}$$

The electric-field direction of circularly polarized light therefore rotates much faster than the response time of the detector.

2.2 Linear analyzer and quarter-wave plate

The unit vector along the transmission axis of a linear analyzer is

$$\mathbf{e}_\theta = \begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix}. \tag{5}$$

The transmitted electric-field amplitude is the projection

$$E_{\mathrm{out}} = \mathbf{e}_\theta^T \mathbf{J}, \tag{6}$$

and the measured intensity is proportional to its squared magnitude:

$$I(\theta) \propto \left| \mathbf{e}_\theta^T \mathbf{J} \right|^2. \tag{7}$$

A quarter-wave plate introduces a relative phase difference of $\pi/2$ between the field components parallel to its fast and slow axes. The corresponding unit vectors are

$$\mathbf{e}_f = \begin{pmatrix} \sin\phi \\ \cos\phi \end{pmatrix}, \qquad \mathbf{e}_s = \begin{pmatrix} \cos\phi \\ -\sin\phi \end{pmatrix}. \tag{8}$$

Apart from an irrelevant global phase, the Jones matrix of the quarter-wave plate may be written as

$$Q(\phi) = \begin{pmatrix} \sin^2\phi + i\cos^2\phi & (1-i)\sin\phi\cos\phi \\ (1-i)\sin\phi\cos\phi & \cos^2\phi + i\sin^2\phi \end{pmatrix}. \tag{9}$$

2.3 Polarization states and analyzer intensity (Task 0)

2.3.1 Linear polarization followed by an analyzer

Using Eqs. (2) and (7), the transmitted amplitude is

$$\begin{aligned} E_{\mathrm{out}} &= \mathbf{e}_\theta^T \mathbf{J}_\alpha && (10)\\ &= \sin\theta\sin\alpha + \cos\theta\cos\alpha && (11)\\ &= \cos(\theta - \alpha). && (12) \end{aligned}$$

The transmitted intensity is therefore

$$I(\theta) = I_0 \cos^2(\theta - \alpha), \tag{13}$$

which is Malus' law. The intensity is maximal when the analyzer is parallel to the incident polarization and ideally vanishes when the two directions differ by $90°$.

2.3.2 Circular polarization followed by an analyzer

For right- or left-circularly polarized light,

$$\mathbf{e}_\theta^T \mathbf{J}_\pm = \frac{1}{\sqrt{2}} \left(\sin\theta \mp i\cos\theta \right). \tag{14}$$

Consequently,

$$\begin{aligned} I_\pm(\theta) &= I_0 \left| \mathbf{e}_\theta^T \mathbf{J}_\pm \right|^2 && (15)\\ &= \frac{I_0}{2} \left(\sin^2\theta + \cos^2\theta \right) && (16)\\ &= \frac{I_0}{2}. && (17) \end{aligned}$$

The analyzer therefore transmits half of the incident intensity independently of its angle. An analyzer alone cannot distinguish circularly polarized light from ideal unpolarized light.

2.3.3 Linear polarization followed by a quarter-wave plate and analyzer

The incident linear polarization can be decomposed along the fast and slow axes:

$$\mathbf{J}_\alpha = \cos(\alpha - \phi)\mathbf{e}_f + \sin(\alpha - \phi)\mathbf{e}_s. \tag{18}$$

After the quarter-wave plate, the two components differ in phase by $\pi/2$:

$$\mathbf{J}_{\mathrm{QWP}} = \cos(\alpha - \phi)\mathbf{e}_f + i\sin(\alpha - \phi)\mathbf{e}_s. \tag{19}$$

Projection onto the analyzer axis gives

$$E_{\mathrm{out}} = \cos(\alpha - \phi)\cos(\theta - \phi) + i\sin(\alpha - \phi)\sin(\theta - \phi). \tag{20}$$

Since the two terms are 90° out of phase, the cross term vanishes when the magnitude is squared:

$$I(\theta) = I_0\left[\cos^2(\alpha - \phi)\cos^2(\theta - \phi) + \sin^2(\alpha - \phi)\sin^2(\theta - \phi)\right]. \tag{21}$$

Using trigonometric identities, this becomes

$$I(\theta) = \frac{I_0}{2}\left[1 + \cos\big(2(\alpha - \phi)\big)\cos\big(2(\theta - \phi)\big)\right]. \tag{22}$$

The special cases requested in Task 0 are:

- For $\alpha = \phi$, the incident polarization is parallel to one principal axis of the plate:

  $$I(\theta) = I_0\cos^2(\theta - \phi). \tag{23}$$

  The polarization remains linear.

- For $\alpha = \phi \pm \pi/2$, the incident polarization is parallel to the other principal axis:

  $$I(\theta) = I_0\sin^2(\theta - \phi). \tag{24}$$

  The polarization also remains linear.

- For $\alpha = \phi \pm \pi$, the polarization direction is physically equivalent to $\alpha = \phi$, since a linear polarization axis has a period of 180°.

- For $\alpha = \phi \pm \pi/4$, both field components have equal amplitudes and acquire a phase difference of $\pm\pi/2$:

  $$I(\theta) = \frac{I_0}{2}. \tag{25}$$

  The outgoing light is circularly polarized. The two signs correspond to opposite handedness.

2.3.4 Circular polarization followed by a quarter-wave plate and analyzer

Applying the quarter-wave-plate matrix to the circular Jones vectors gives, up to an irrelevant global phase,

$$Q(\phi)\mathbf{J}_+ \propto \mathbf{J}_{\phi-\pi/4}, \qquad Q(\phi)\mathbf{J}_- \propto \mathbf{J}_{\phi+\pi/4}. \tag{26}$$

Circularly polarized light is therefore converted into linearly polarized light. The analyzer intensities are

$$I_+(\theta) = I_0\cos^2\left[\theta - \left(\phi - \frac{\pi}{4}\right)\right], \tag{27}$$

and

$$I_-(\theta) = I_0\cos^2\left[\theta - \left(\phi + \frac{\pi}{4}\right)\right]. \tag{28}$$

For $\phi = +\pi/4$, right-circularly polarized light becomes vertically linearly polarized, while left-circularly polarized light becomes horizontally linearly polarized. For $\phi = -\pi/4$, the two results are interchanged.

2.4 Expected intensity curves for Task 1

In the experiment, the analyzer position defined as $\theta = 0°$ was adjusted to give nearly complete extinction of the incident linearly polarized light. The incident polarization direction was therefore approximately

$$\alpha = 90°. \tag{29}$$

Substitution into Eq. (22) gives

$$I(\theta, \phi) = \frac{I_0}{2}\left[1 - \cos(2\phi)\cos\big(2(\theta - \phi)\big)\right]. \tag{30}$$

For the three plate angles used in Task 1, the theoretical predictions are:

$\phi = 0°$**:**

$$I(\theta) = I_0 \sin^2 \theta. \tag{31}$$

The intensity is minimal at 0° and 180°, and maximal at 90°.

$\phi = 30°$**:**

$$I(\theta) = \frac{I_0}{2}\left[1 - \frac{1}{2}\cos\big(2(\theta - 30°)\big)\right]. \tag{32}$$

The expected extrema are

$$I_{\min} = \frac{I_0}{4}, \qquad I_{\max} = \frac{3I_0}{4}. \tag{33}$$

The resulting state is elliptically polarized.

$\phi = 45°$**:**

$$I(\theta) = \frac{I_0}{2}. \tag{34}$$

The ideal output is circularly polarized and independent of analyzer angle.

The quarter-wave plate transmits approximately 90% of the incident intensity. This reduces the absolute detector signal but does not change the ideal angular modulation within an individual scan.

2.5 General polarization state and Stokes parameters

A general fully polarized state may be written as

$$\mathbf{E} = \begin{pmatrix} E_{0x} \\ E_{0y} e^{-i\delta} \end{pmatrix}, \tag{35}$$

where δ is the phase difference between the two orthogonal field components. The state is linear for $\delta = m\pi$, circular when $E_{0x} = E_{0y}$ and $\delta = \pm\pi/2$, and elliptical in the general case.

Projection onto a linear analyzer gives

$$I(\theta) \propto \left|E_{0x} \sin\theta + E_{0y} e^{-i\delta} \cos\theta\right|^2 \tag{36}$$

$$= E_{0x}^2 \sin^2\theta + E_{0y}^2 \cos^2\theta + 2E_{0x}E_{0y} \cos\delta \sin\theta \cos\theta. \tag{37}$$

The Stokes parameters are defined as

$$s_0 = E_{0x}^2 + E_{0y}^2, \tag{38}$$

$$s_1 = E_{0x}^2 - E_{0y}^2, \tag{39}$$

$$s_2 = 2E_{0x}E_{0y} \cos\delta, \tag{40}$$

$$s_3 = 2E_{0x}E_{0y} \sin\delta. \tag{41}$$

Equation (37) can then be written as

$$I(\theta) = \frac{1}{2}\left[s_0 - s_1 \cos(2\theta) + s_2 \sin(2\theta)\right]. \tag{42}$$

A linear analyzer is therefore sensitive to s_1 and s_2, but not directly to s_3. The circular component must be converted into a linear component using a quarter-wave plate.

The Stokes parameters may be determined from six intensity measurements:

$$s_0 \propto I_{\pi/2} + I_0, \tag{43}$$

$$s_1 \propto I_{\pi/2} - I_0, \tag{44}$$

$$s_2 \propto I_{\pi/4} - I_{-\pi/4}, \tag{45}$$

$$s_3 \propto I_+ - I_-. \tag{46}$$

For fully polarized light,

$$s_0^2 = s_1^2 + s_2^2 + s_3^2. \tag{47}$$

The degree of polarization is

$$P = \frac{\sqrt{s_1^2 + s_2^2 + s_3^2}}{s_0}, \tag{48}$$

where $P = 1$ corresponds to fully polarized light and $P < 1$ indicates an unpolarized contribution.

2.6 Operational classification used in Task 2

The unknown polarization states can also be identified without determining all Stokes parameters.

- If the analyzer-only signal is independent of angle, the light is either circularly polarized or unpolarized. After insertion of a quarter-wave plate:
  - no angular variation indicates unpolarized light;
  - a variation with a zero minimum indicates circularly polarized light;
  - a variation with a nonzero minimum indicates a mixture of circularly polarized and unpolarized light.
- If the analyzer-only signal varies with angle:
  - a minimum consistent with zero indicates linearly polarized light;
  - a nonzero minimum may indicate elliptical polarization or a polarized–unpolarized mixture.
- For an analyzer-only curve with a nonzero minimum, the quarter-wave plate is aligned with a principal direction of the measured curve:
  - if a zero minimum then becomes possible, the original light was elliptically polarized;
  - if the minimum remains nonzero at approximately the same analyzer angle, the light is a mixture of linearly polarized and unpolarized light;
  - if the minimum remains nonzero but shifts to another analyzer angle, the light is a mixture of elliptically polarized and unpolarized light.

2.7 Relation between detector voltage and intensity

The photodiode voltage is proportional to the incident optical intensity. The blocked-LED background voltage was

$$U_{\mathrm{bg}} = 0.0016\,\mathrm{V}, \tag{49}$$

so the background-corrected detector signal is

$$S(\theta) = U(\theta) - U_{\mathrm{bg}} \propto I(\theta). \tag{50}$$

The measured analyzer curves can be fitted using

$$S(\theta) = a + b\cos(2\theta) + c\sin(2\theta), \tag{51}$$

or equivalently

$$S(\theta) = a + R\cos\left[2(\theta - \theta_0)\right], \tag{52}$$

where

$$R = \sqrt{b^2 + c^2}. \tag{53}$$

The maximum and minimum fitted signals are

$$S_{\max} = a + R, \qquad S_{\min} = a - R. \tag{54}$$

The modulation depth is therefore

$$C = \frac{R}{a} = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}}. \tag{55}$$

For ideal linearly polarized light $C = 1$, while an analyzer-independent signal has $C = 0$.

3 Experimental Analysis

3.1 Experimental Setup

The optical setup consisted of a yellow LED, a known linear polarizer or one of three unknown black boxes, a rotatable quarter-wave plate, a rotatable linear analyzer, a collimating lens, and a photodiode detector. The components were arranged as

$$\text{LED} \rightarrow \text{polarizer/black box} \rightarrow \lambda/4\text{-plate} \rightarrow \text{analyzer} \rightarrow \text{lens} \rightarrow \text{photodiode}. \tag{56}$$

The LED had a nominal wavelength of

$$\lambda = 590\,\mathrm{nm}, \tag{57}$$

and the photodiode circuit was operated using a supply voltage of approximately

$$V_{\mathrm{supply}} \approx 5\,\mathrm{V}. \tag{58}$$

The voltage measured across the detector circuit was taken to be proportional to the incident optical intensity.

The voltage recorded with the LED beam blocked was

$$U_{\mathrm{bg}} = 0.0016\,\mathrm{V}. \tag{59}$$

The background-corrected detector signal was therefore calculated as

$$S = U - U_{\mathrm{bg}}. \tag{60}$$

According to an estimate provided by the supervisor, insertion of the quarter-wave plate reduced the transmitted intensity by approximately 10%. This value was treated as an approximate systematic effect rather than as an independently measured quantity.

3.2 Measurement Uncertainties

The analyzer scale had a smallest division of 1°, and its angular uncertainty was taken as

$$\sigma_\theta = 1^\circ. \tag{61}$$

The uncertainty of the quarter-wave-plate orientation was taken from the experiment manual as

$$\sigma_\phi = 2^\circ. \tag{62}$$

The final two digits of the measured detector voltage fluctuated slightly. Since repeated voltage measurements were not recorded at every angle, the voltage uncertainty was estimated from the observed fluctuations and from the scatter of the data around the fitted curves.

Further systematic uncertainties arose from manual angular adjustment, possible scale-reading parallax, slow LED intensity drift, imperfect polarizer extinction, optical misalignment, and deviations of the quarter-wave-plate retardance from its ideal value of $\pi/2$.

3.3 Data Evaluation

For each measurement, the background-corrected signal from Eq. (60) was fitted using

$$S(\theta) = a + b\cos(2\theta) + c\sin(2\theta), \tag{63}$$

or equivalently

$$S(\theta) = a + R\cos\left[2(\theta - \theta_0)\right], \tag{64}$$

with

$$R = \sqrt{b^2 + c^2}. \tag{65}$$

The fitted maximum and minimum signals were

$$S_{\max} = a + R, \qquad S_{\min} = a - R, \tag{66}$$

and the modulation depth was calculated as

$$C = \frac{R}{a} = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}}. \tag{67}$$

4 Polarization after a Quarter-Wave Plate (Task 1)

4.1 Goal

The aim of this task was to investigate the polarization state produced when initially linearly polarized light passed through a quarter-wave plate at the orientations

$$\phi = 0^\circ, \qquad 30^\circ, \qquad 45^\circ. \tag{68}$$

For each plate orientation, the detector voltage was measured as a function of the analyzer angle and compared with the theoretical prediction derived from Jones calculus.

The analyzer position defined as $\theta = 0^\circ$ was adjusted to give approximately minimum transmission before the measurements. The incident linear polarization was therefore approximately perpendicular to this analyzer direction, corresponding to

$$\alpha \approx 90^\circ. \tag{69}$$

4.2 Measurement Procedure

The known linear polarizer was fixed directly in front of the LED. The quarter-wave plate was initially set to $\phi = 0°$, and the input polarizer was adjusted until the detector voltage was nearly minimal at the analyzer position defined as $\theta = 0°$.

For each quarter-wave-plate orientation, the analyzer was rotated over the interval

$$0° \leq \theta \leq 180° \tag{70}$$

in increments of 10°. This resulted in 19 voltage measurements for each of the three plate settings.

The voltage measured with the LED beam blocked was

$$U_{\mathrm{bg}} = 0.0016\,\mathrm{V}. \tag{71}$$

The background-corrected signal was therefore

$$S(\theta) = U(\theta) - U_{\mathrm{bg}}. \tag{72}$$

The analyzer and quarter-wave-plate uncertainties were taken as

$$\sigma_\theta = 1°, \qquad \sigma_\phi = 2°. \tag{73}$$

4.3 Theoretical Model

For the experimental convention $\alpha = 90°$, the expected analyzer dependence is

$$S_{\mathrm{th}}(\theta, \phi) = \frac{S_0}{2}\left[1 - \cos(2\phi)\cos\left(2(\theta - \phi)\right)\right], \tag{74}$$

where S_0 is the signal corresponding to the maximum intensity of the incident linearly polarized light.

For $\phi = 0°$,

$$S_{\mathrm{th}}(\theta, 0°) = S_0 \sin^2\theta. \tag{75}$$

The output remains linearly polarized, with a minimum near 0° and a maximum near 90°.
For $\phi = 30°$,

$$S_{\mathrm{th}}(\theta, 30°) = \frac{S_0}{2}\left[1 - \frac{1}{2}\cos\left(2(\theta - 30°)\right)\right]. \tag{76}$$

The expected extrema are

$$S_{\mathrm{min}} = \frac{S_0}{4}, \qquad S_{\mathrm{max}} = \frac{3S_0}{4}, \tag{77}$$

and the output is elliptically polarized.
For $\phi = 45°$,

$$S_{\mathrm{th}}(\theta, 45°) = \frac{S_0}{2}. \tag{78}$$

The analyzer signal should ideally be independent of angle, corresponding to circular polarization.

The approximate 10% absorption of the quarter-wave plate affects the absolute voltage scale but not the predicted modulation within an individual analyzer scan.

4.4 Fit Model and Data Evaluation

Each background-corrected dataset was fitted using

$$S(\theta) = a + b\cos(2\theta) + c\sin(2\theta), \tag{79}$$

which is equivalent to

$$S(\theta) = a + R\cos\left[2(\theta - \theta_{\mathrm{max}})\right]. \tag{80}$$

The modulation amplitude is

$$R = \sqrt{b^2 + c^2}, \tag{81}$$

and the fitted extrema are

$$S_{\mathrm{max}} = a + R, \qquad S_{\mathrm{min}} = a - R. \tag{82}$$

The modulation depth was calculated as

$$C = \frac{R}{a} = \frac{S_{\mathrm{max}} - S_{\mathrm{min}}}{S_{\mathrm{max}} + S_{\mathrm{min}}}. \tag{83}$$

For ideal linear polarization, $C = 1$, whereas for an analyzer-independent signal, $C = 0$.

Because repeated voltage readings were not recorded, the point-to-point voltage scatter was estimated from the fit residuals:

$$\sigma_{U,\mathrm{emp}} = \sqrt{\frac{\sum_{i=1}^{N} \left[S_i - S_{\mathrm{fit}}(\theta_i)\right]^2}{N - p}}, \tag{84}$$

where $N = 19$ is the number of measurements and $p = 3$ is the number of fitted parameters.

4.5 Results

Figure 1 presents the three uncorrected datasets together. The transition from a strongly modulated curve at $\phi = 0°$, through a curve with reduced modulation at $\phi = 30°$, to an almost angle-independent signal at $\phi = 45°$ is clearly visible.

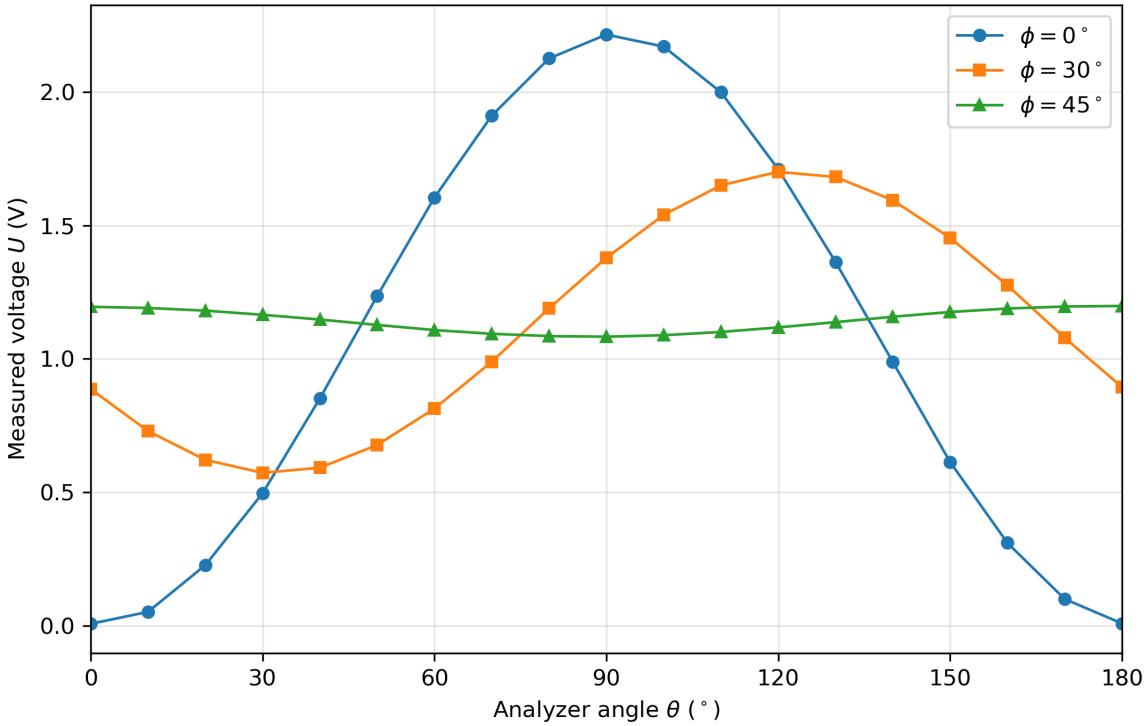


Figure 1: Raw voltages as a function of analyzer angle for the three quarter-wave-plate orientations.

The numerical fit results are summarized in Table 1. The uncertainties stated for C are statistical fit uncertainties only. The experimentally relevant angular uncertainty is $\sigma_\theta = 1°$.

Table 1: Fit results for the three quarter-wave-plate orientations. The signals are background-corrected. The uncertainty of C is obtained from the fit covariance matrix.

ϕ	C_{exp}	C_{th}	θ_{max}	$\theta_{\mathrm{max,th}}$	S_{min} (V)	S_{max} (V)	R^2
0°	0.9965 ± 0.0012	1.000	$(91.7 \pm 1.0)°$	90°	0.00385	2.21323	0.999985
30°	0.4992 ± 0.0007	0.500	$(122.2 \pm 1.0)°$	120°	0.56751	1.69905	0.999971
45°	0.0501 ± 0.0004	0.000	–	–	1.08239	1.19652	0.999207

The individual measurements, fitted curves, and ideal theoretical predictions are shown in Figs. 2–4.

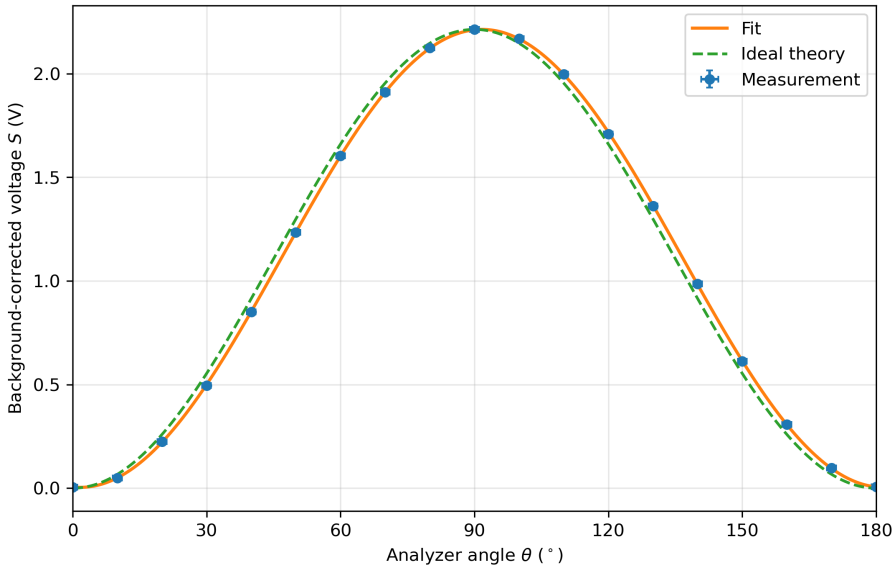


Figure 2: Background-corrected detector signal for $\phi = 0°$.

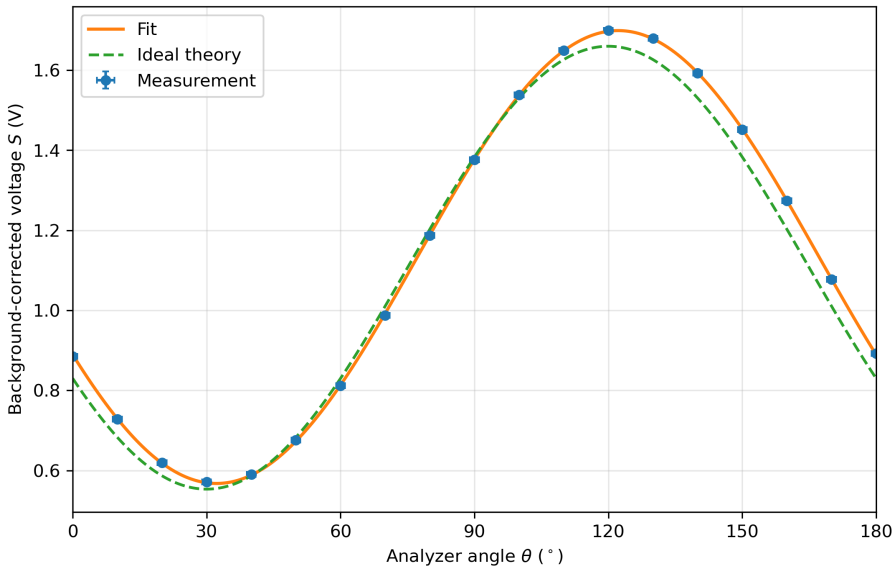


Figure 3: Background-corrected detector signal for $\phi = 30°$.

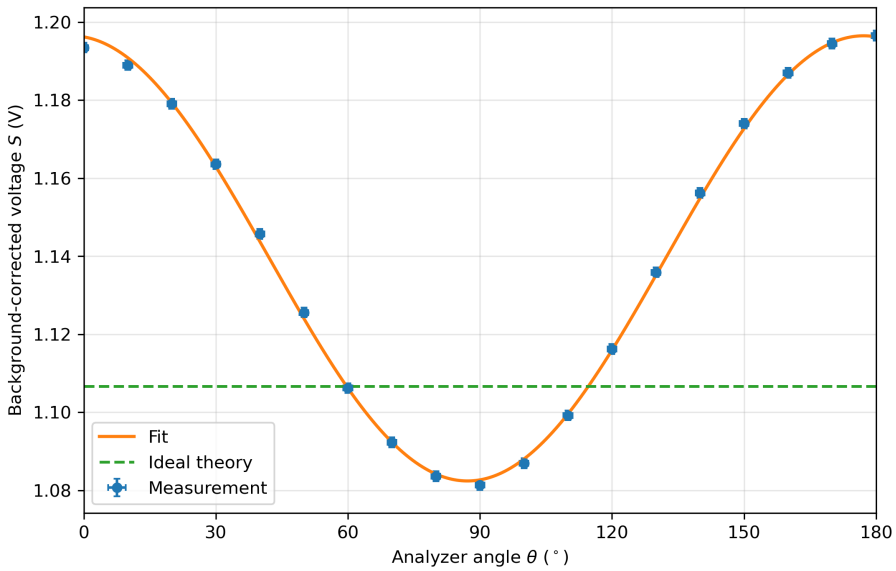


Figure 4: Background-corrected detector signal for $\phi = 45°$.

Using the fitted maximum of the $\phi = 0^\circ$ measurement as the reference signal S_0, the experimentally determined intensity ratios were

$$\frac{S_{\min,30}}{S_0} = 0.2564, \qquad \left(\frac{S_{\min,30}}{S_0}\right)_{\text{th}} = 0.2500, \tag{85}$$

$$\frac{S_{\max,30}}{S_0} = 0.7677, \qquad \left(\frac{S_{\max,30}}{S_0}\right)_{\text{th}} = 0.7500, \tag{86}$$

$$\frac{\overline{S}_{45}}{S_0} = 0.5148, \qquad \left(\frac{\overline{S}_{45}}{S_0}\right)_{\text{th}} = 0.5000. \tag{87}$$

4.6 Uncertainty Analysis

The empirical voltage uncertainties obtained from the residual scatter were

$$\sigma_{U,\text{emp}} = \begin{cases} 0.00340\,\text{V}, & \phi = 0^\circ, \\ 0.00230\,\text{V}, & \phi = 30^\circ, \\ 0.00127\,\text{V}, & \phi = 45^\circ. \end{cases} \tag{88}$$

These values represent the observed point-to-point scatter and should not be interpreted as manufacturer-specified multimeter uncertainties.

The fit returns very small formal angular uncertainties, for example approximately 0.03° for the first two measurements. These values do not include manual scale reading, alignment, or calibration errors and are therefore not representative of the total experimental uncertainty. The analyzer uncertainty of 1° was used when reporting the measured maximum angles.

For comparison with the theoretical maximum positions, both the analyzer and QWP uncertainties are relevant. Assuming independent uncertainties,

$$\sigma_{\Delta\theta} = \sqrt{\sigma_\theta^2 + \sigma_\phi^2} = \sqrt{(1^\circ)^2 + (2^\circ)^2} \approx 2.2^\circ. \tag{89}$$

The deviations

$$91.7^\circ - 90.0^\circ = 1.7^\circ \tag{90}$$

and

$$122.2^\circ - 120.0^\circ = 2.2^\circ \tag{91}$$

are therefore compatible with the theoretical predictions.

The relative uncertainty depends strongly on the measured signal. Near the minimum of the $\phi = 0^\circ$ scan,

$$\frac{\sigma_{U,\text{emp}}}{S_{\min}} \approx \frac{0.00340}{0.00385} \approx 0.88. \tag{92}$$

Thus, the relative uncertainty near extinction is of order 90%. At the maximum of the same curve,

$$\frac{\sigma_{U,\text{emp}}}{S_{\max}} \approx \frac{0.00340}{2.213} \approx 1.5 \times 10^{-3}, \tag{93}$$

or approximately 0.15%. The largest relative error therefore occurs close to the intensity minima, where background subtraction and voltage fluctuations are comparable to the remaining signal.

The principal systematic uncertainties were the angular alignment of the analyzer and QWP, imperfect polarizer extinction, deviations of the QWP retardance from exactly $\pi/2$, the finite spectral width of the LED, and possible slow drift of the LED intensity.

4.7 Interpretation and Discussion

4.7.1 $\phi = 0^\circ$

The measured curve shows an almost ideal sinusoidal dependence with a minimum close to $\theta = 0^\circ$ and a maximum at

$$\theta_{\max} = (91.7 \pm 1.0)^\circ. \tag{94}$$

The modulation depth

$$C = 0.9965 \pm 0.0012 \tag{95}$$

is close to the ideal value $C = 1$. The light therefore remained almost completely linearly polarized after passing through the QWP with its axis aligned with the incident polarization.

The small nonzero fitted minimum is comparable to the empirical voltage uncertainty. It can therefore be explained by detector fluctuations, residual background light, finite extinction of the polarizers, and slight angular misalignment.

4.7.2 $\phi = 30°$

For $\phi = 30°$, the fitted modulation depth was

$$C = 0.4992 \pm 0.0007, \tag{96}$$

in excellent agreement with the theoretical value $C = 0.5$. The maximum occurred at

$$\theta_{\max} = (122.2 \pm 1.0)°, \tag{97}$$

which agrees with the expected 120° within the combined angular uncertainty.

The intensity remained nonzero for every analyzer angle, and the measured extrema were close to the theoretical values $S_0/4$ and $3S_0/4$. The output was therefore elliptically polarized.

4.7.3 $\phi = 45°$

For an ideal QWP at $\phi = 45°$, the output should be circularly polarized and the analyzer signal should be constant. The measured mean intensity ratio,

$$\frac{\overline{S}_{45}}{S_0} = 0.5148, \tag{98}$$

was close to the theoretical value 0.5.

A small but clearly resolved modulation remained:

$$C = 0.0501 \pm 0.0004. \tag{99}$$

Consequently, the output was approximately circularly polarized but retained a small elliptical component.

The residual modulation is compatible with the QWP angular uncertainty. For a deviation of 2° from the ideal setting,

$$C_{\text{th}} = |\cos(2(45° \pm 2°))| \approx 0.070. \tag{100}$$

The observed value $C \approx 0.050$ therefore lies within the range that can be produced by the stated plate-angle uncertainty alone. Additional contributions may arise from nonideal retardance at 590 nm, the spectral width of the LED, imperfect input polarization, and optical misalignment.

The fitted maximum angle for this measurement was not interpreted physically. For an ideal circular state, all analyzer angles are equivalent, so the maximum direction is undefined. The fit assigns a maximum only because of the small residual modulation.

4.8 Conclusion

The measurements showed the expected progression from linear through elliptical to approximately circular polarization as the QWP angle was increased from 0° to 45°. The fitted modulation depths for $\phi = 0°$ and 30° agreed closely with the theoretical values, while the $\phi = 45°$ measurement showed a small residual modulation consistent with the angular uncertainty and nonideal optical components.

Overall, the data were well described by the theoretical analyzer model, with all fits yielding

$$R^2 > 0.999. \tag{101}$$

The dominant relative uncertainty occurred near complete extinction, whereas the dominant systematic uncertainty in the comparison with theory was the QWP angular alignment.

5 Characterization of the Unknown Polarization States (Task 2)

5.1 Goal

The aim of Task 2 was to determine the polarization states produced by the three unknown optical elements BBB1, BBB2, and BBB3. Each state was first examined using a rotatable linear analyzer. Where the analyzer-only measurement did not provide an unambiguous classification, a quarter-wave plate was inserted and a second complete analyzer scan was recorded.

The classification followed the operational procedure

$$\text{measurement} \rightarrow \text{interpretation} \rightarrow \text{additional measurement} \rightarrow \text{final conclusion}. \tag{102}$$

Complete angular scans were used instead of individual measurement points so that the conclusions did not depend on a single possibly inaccurate reading.

5.2 Measurement and Evaluation Method

For each black box, the analyzer was rotated from the right-hand position

$$\theta = -90^\circ \tag{103}$$

to the left-hand position

$$\theta = +90^\circ \tag{104}$$

in increments of 10°. This resulted in 19 measurement points per scan.

The first scan was performed without the quarter-wave plate. The analyzer angle corresponding to the largest measured voltage was then identified. For the second scan, the quarter-wave plate was inserted with its axis set approximately to this maximum-intensity direction. The settings were

$$\phi_{\mathrm{BBB1}} = 80^\circ, \qquad \phi_{\mathrm{BBB2}} = 0^\circ, \qquad \phi_{\mathrm{BBB3}} = 60^\circ. \tag{105}$$

The measured blocked-LED background voltage was subtracted according to

$$S(\theta) = U(\theta) - U_{\mathrm{bg}}, \qquad U_{\mathrm{bg}} = 0.0016\,\mathrm{V}. \tag{106}$$

The analyzer-angle dependence was fitted using

$$S(\theta) = a + b\cos(2\theta) + c\sin(2\theta), \tag{107}$$

or equivalently

$$S(\theta) = a + R\cos\left[2(\theta - \theta_{\max})\right], \tag{108}$$

where

$$R = \sqrt{b^2 + c^2}. \tag{109}$$

The fitted maximum and minimum signals were

$$S_{\max} = a + R, \qquad S_{\min} = a - R. \tag{110}$$

The modulation depth was calculated as

$$C = \frac{R}{a} = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}}, \tag{111}$$

and the relative minimum was characterized by

$$r_{\min} = \frac{S_{\min}}{S_{\max}}. \tag{112}$$

For ideal linearly polarized light,

$$C = 1, \qquad r_{\min} = 0. \tag{113}$$

A nonzero modulation together with a large nonzero minimum indicates either elliptical polarization or a mixture of polarized and unpolarized light. The effect of the quarter-wave plate is then needed to distinguish between these possibilities.

Because the average of the analyzer signal over a full 180° rotation is represented by the fit parameter a, the approximate transmission of the QWP was estimated using

$$T_{\mathrm{QWP}} \approx \frac{a_{\text{with QWP}}}{a_{\text{analyzer only}}}, \tag{114}$$

with the corresponding loss

$$L_{\mathrm{QWP}} = (1 - T_{\mathrm{QWP}})\,100\%. \tag{115}$$

5.3 Classification Criteria

The following criteria were used:

- A strongly modulated analyzer-only signal with a minimum consistent with zero indicates linearly polarized light.
- An analyzer-dependent signal with a large nonzero minimum is not sufficient by itself to distinguish fully elliptical polarization from a mixture containing an unpolarized contribution.
- If insertion of the QWP produces a nearly zero minimum, the original light was fully elliptically polarized because the QWP transformed it into an approximately linear state.
- If the minimum remains nonzero and the modulation remains approximately unchanged after insertion of the QWP, the light is consistent with a mixture of linearly polarized and unpolarized light.
- If the analyzer-only signal were constant, additional QWP measurements would be required to distinguish circularly polarized light from unpolarized light. This case did not occur for the three investigated boxes.

5.4 Results

The numerical fit results are summarized in Table 2. The quoted angular uncertainty is the experimental analyzer uncertainty

$$\sigma_\theta = 1^\circ. \tag{116}$$

The formal fit uncertainties were substantially smaller but did not include manual positioning and systematic alignment effects. The QWP uncertainty was

$$\sigma_\phi = 2^\circ. \tag{117}$$

Table 2: Fit results for the analyzer-only and QWP measurements of the three black boxes. All signals are corrected for the blocked-LED background voltage.

Box	Configuration	C	$\theta_{\max}$	$S_{\min}$ (V)	$S_{\max}$ (V)	$S_{\min}/S_{\max}$	R^2
BBB1	Analyzer only	0.1906	$(84.1 \pm 1.0)^\circ$	1.95349	2.87348	0.67984	0.999843
BBB1	QWP, $\phi = 80^\circ$	0.1876	$(79.1 \pm 1.0)^\circ$	1.76140	2.57481	0.68409	0.999881
BBB2	Analyzer only	0.9990	$(-4.6 \pm 1.0)^\circ$	0.000687	1.43603	0.000478	0.999988
BBB2	QWP, $\phi = 0^\circ$	0.9803	$(-1.1 \pm 1.0)^\circ$	0.01282	1.28553	0.00997	0.999979
BBB3	Analyzer only	0.2132	$(60.3 \pm 1.0)^\circ$	0.88978	1.37210	0.64848	0.999968
BBB3	QWP, $\phi = 60^\circ$	0.9902	$(-82.4 \pm 1.0)^\circ$	0.00969	1.96383	0.00493	0.999990

A compact comparison of the QWP effect and the resulting classifications is given in Table 3.

Table 3: Effect of the QWP and final classification of each black box. The QWP loss was calculated from the ratio of the fitted mean signals.

Box	C_{without}	C_{with}	T_{QWP}	QWP loss	Classification
BBB1	0.1906	0.1876	0.898	10.2%	Linear + unpolarized mixture
BBB2	0.9990	0.9803	0.904	9.6%	Linear polarization
BBB3	0.2132	0.9902	0.873	12.7%	Elliptical polarization

5.5 BBB1

5.5.1 Measurement without the QWP

The analyzer-only measurement for BBB1 showed a clear but relatively weak angular modulation, as shown in Fig. 5. The fitted modulation depth was

$$C_{\text{without}} = 0.1906, \tag{118}$$

and the fitted minimum-to-maximum ratio was

$$\frac{S_{\min}}{S_{\max}} = 0.6798. \tag{119}$$

The minimum was therefore far above zero. This excludes fully linearly polarized light.

The maximum occurred near

$$\theta_{\max} = (84.1 \pm 1.0)^\circ. \tag{120}$$

The QWP was consequently set to the nearest observed maximum direction,

$$\phi = 80^\circ. \tag{121}$$

5.5.2 Measurement with the QWP

After insertion of the QWP, the modulation depth became

$$C_{\text{with}} = 0.1876, \tag{122}$$

while the minimum-to-maximum ratio was

$$\frac{S_{\min}}{S_{\max}} = 0.6841. \tag{123}$$

Both quantities remained almost unchanged. The minimum also remained large and occurred at approximately the same analyzer direction.

The fitted mean signal decreased by the factor

$$T_{\mathrm{QWP}} = \frac{2.16810}{2.41349} = 0.898, \tag{124}$$

corresponding to a loss of approximately

$$L_{\mathrm{QWP}} \approx 10.2\%. \tag{125}$$

This agrees closely with the supervisor's estimate that the QWP absorbs approximately 10%. The principal effect of inserting the QWP was therefore a reduction in total signal rather than a change in polarization behavior.

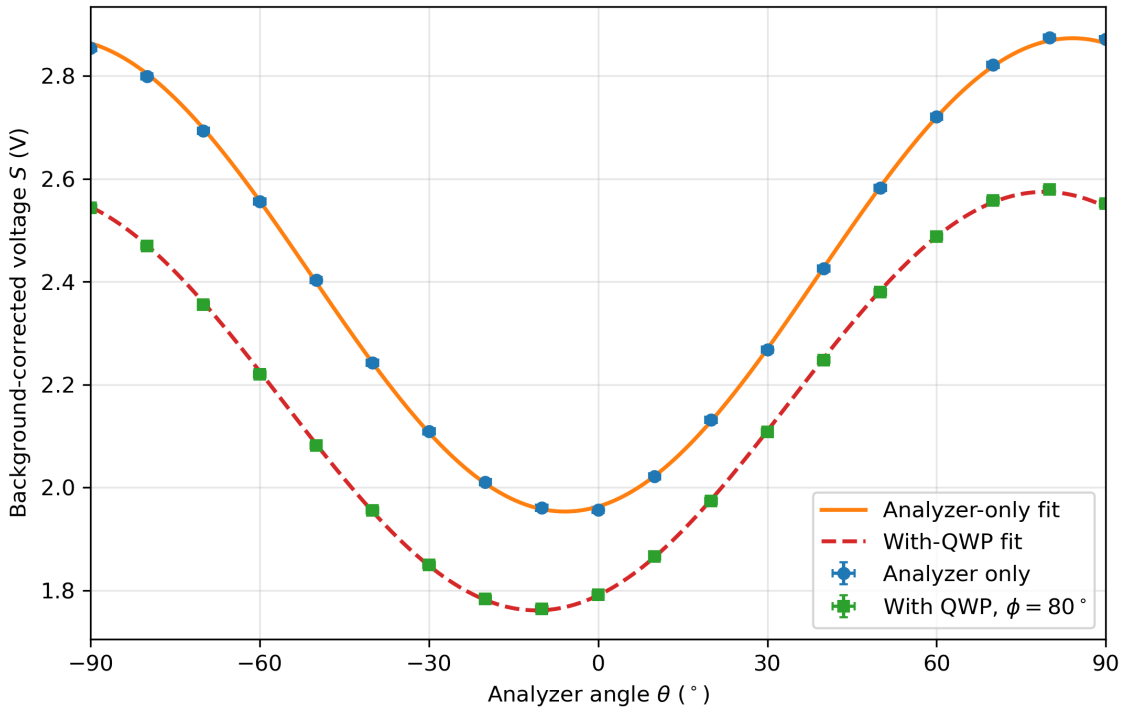


Figure 5: Background-corrected analyzer scans for BBB1 without and with the QWP at $\phi = 80°$.

5.5.3 Reasoning and Classification

The reasoning chain was

$$\begin{aligned} &\text{analyzer dependence with a large nonzero minimum} \\ &\quad \rightarrow \text{polarized component plus an angle-independent component} \\ &\quad \rightarrow \text{QWP insertion leaves the relative modulation unchanged} \\ &\quad \rightarrow \boxed{\text{mixture of linearly polarized and unpolarized light}}. \end{aligned} \tag{126}$$

Within a simple model consisting of a completely linearly polarized component and an unpolarized component, the modulation depth suggests a polarized fraction of approximately

$$P_{\mathrm{lin}} \approx C \approx 0.19. \tag{127}$$

This is only an estimate because a complete Stokes-parameter measurement was not performed and nonideal optical components may also affect the modulation.

5.6 BBB2

5.6.1 Measurement without the QWP

The analyzer-only measurement for BBB2 showed an almost complete $\cos^2$-type modulation, as shown in Fig. 6. The modulation depth was

$$C_{\mathrm{without}} = 0.9990. \tag{128}$$

The fitted extrema were

$$S_{\mathrm{min}} = 0.000687\,\mathrm{V}, \qquad S_{\mathrm{max}} = 1.43603\,\mathrm{V}. \tag{129}$$

Thus,

$$\frac{S_{\mathrm{min}}}{S_{\mathrm{max}}} = 4.78 \times 10^{-4}. \tag{130}$$

The corrected minimum was therefore effectively zero compared with the maximum, providing direct evidence for linearly polarized light.

The fitted maximum occurred at

$$\theta_{\max} = (-4.6 \pm 1.0)^\circ. \tag{131}$$

The QWP was set to the nearest experimentally observed maximum direction,

$$\phi = 0^\circ. \tag{132}$$

5.6.2 Measurement with the QWP

With the QWP inserted parallel to the approximate polarization direction, the signal remained strongly modulated:

$$C_{\text{with}} = 0.9803. \tag{133}$$

The fitted relative minimum increased slightly to

$$\frac{S_{\min}}{S_{\max}} = 0.00997, \tag{134}$$

but remained close to zero.
The mean-signal ratio was

$$T_{\text{QWP}} = \frac{0.64917}{0.71836} = 0.904, \tag{135}$$

corresponding to a loss of approximately

$$L_{\text{QWP}} \approx 9.6\%. \tag{136}$$

This is consistent with the expected QWP absorption.

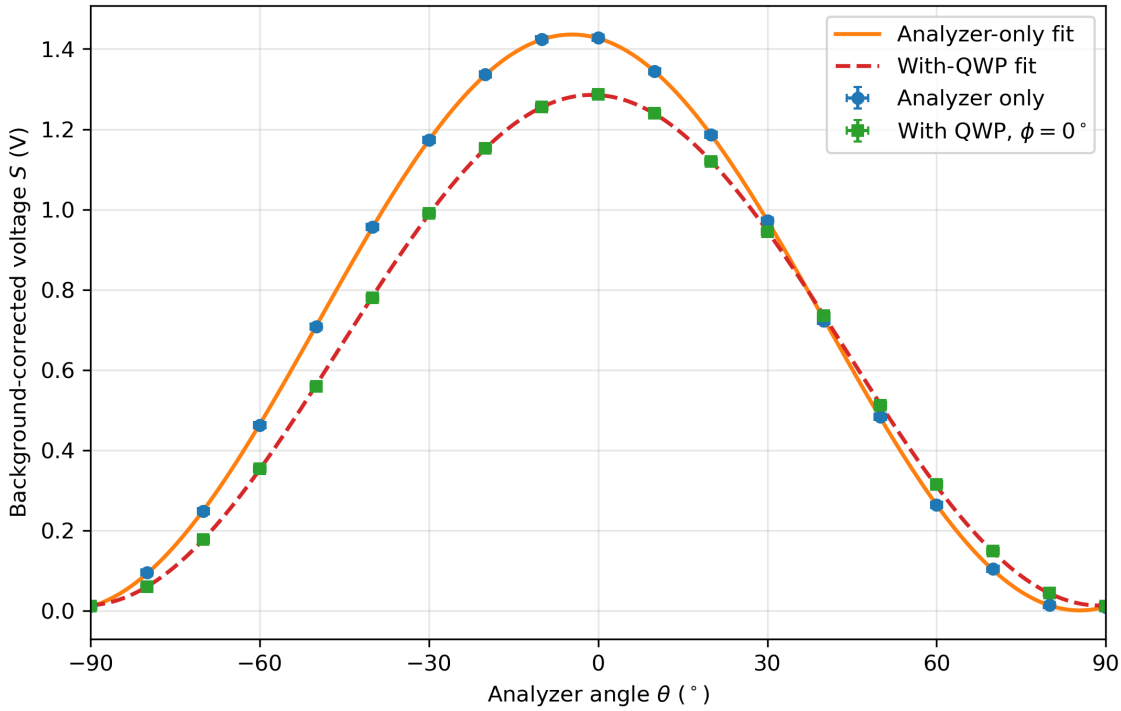


Figure 6: Background-corrected analyzer scans for BBB2 without and with the QWP at $\phi = 0^\circ$.

5.6.3 Reasoning and Classification

The reasoning chain was

$$\begin{aligned} &\text{strong analyzer modulation} \\ &\quad \rightarrow \text{minimum approximately equal to the corrected zero level} \\ &\quad \rightarrow \text{nearly complete extinction is possible} \\ &\quad \rightarrow \boxed{\text{linearly polarized light}}. \end{aligned} \tag{137}$$

The measurement with the QWP provided an additional consistency check. Alignment of a QWP axis with a linear polarization direction should preserve the linear character, apart from absorption and small imperfections. This behavior was observed.

The small nonzero minimum after insertion of the QWP may be caused by finite polarizer extinction, imperfect QWP alignment, nonideal retardance, residual detector background, or slight movement of the beam when the plate was inserted.

5.7 BBB3

5.7.1 Measurement without the QWP

The analyzer-only scan for BBB3 showed a clear modulation but a large nonzero minimum, as shown in Fig. 7. The fitted modulation depth was

$$C_{\text{without}} = 0.2132, \tag{138}$$

with

$$S_{\min} = 0.88978\,\text{V}, \qquad S_{\max} = 1.37210\,\text{V}. \tag{139}$$

The relative minimum was

$$\frac{S_{\min}}{S_{\max}} = 0.6485. \tag{140}$$

The state was therefore not linearly polarized. However, the analyzer-only measurement could not distinguish between fully elliptical polarization and a partially polarized mixture.

The fitted maximum occurred at

$$\theta_{\max} = (60.3 \pm 1.0)^\circ. \tag{141}$$

The QWP was consequently set to

$$\phi = 60^\circ. \tag{142}$$

5.7.2 Measurement with the QWP

After inserting the QWP, the analyzer dependence changed strongly. The modulation depth increased to

$$C_{\text{with}} = 0.9902, \tag{143}$$

and the fitted relative minimum decreased to

$$\frac{S_{\min}}{S_{\max}} = 0.00493. \tag{144}$$

The corrected minimum was

$$S_{\min} = 0.00969\,\text{V}, \tag{145}$$

compared with

$$S_{\max} = 1.96383\,\text{V}. \tag{146}$$

The light could therefore be almost completely extinguished after passing through the QWP. This shows that the QWP converted the initial state into an approximately linearly polarized state.

The mean-signal ratio was

$$T_{\text{QWP}} = \frac{0.98676}{1.13094} = 0.873, \tag{147}$$

which corresponds to a loss of approximately

$$L_{\text{QWP}} \approx 12.7\%. \tag{148}$$

Although this is somewhat larger than the approximate 10% supervisor estimate, the difference is reasonable because the estimate was not a calibrated value and the comparison may also include source drift and alignment changes.

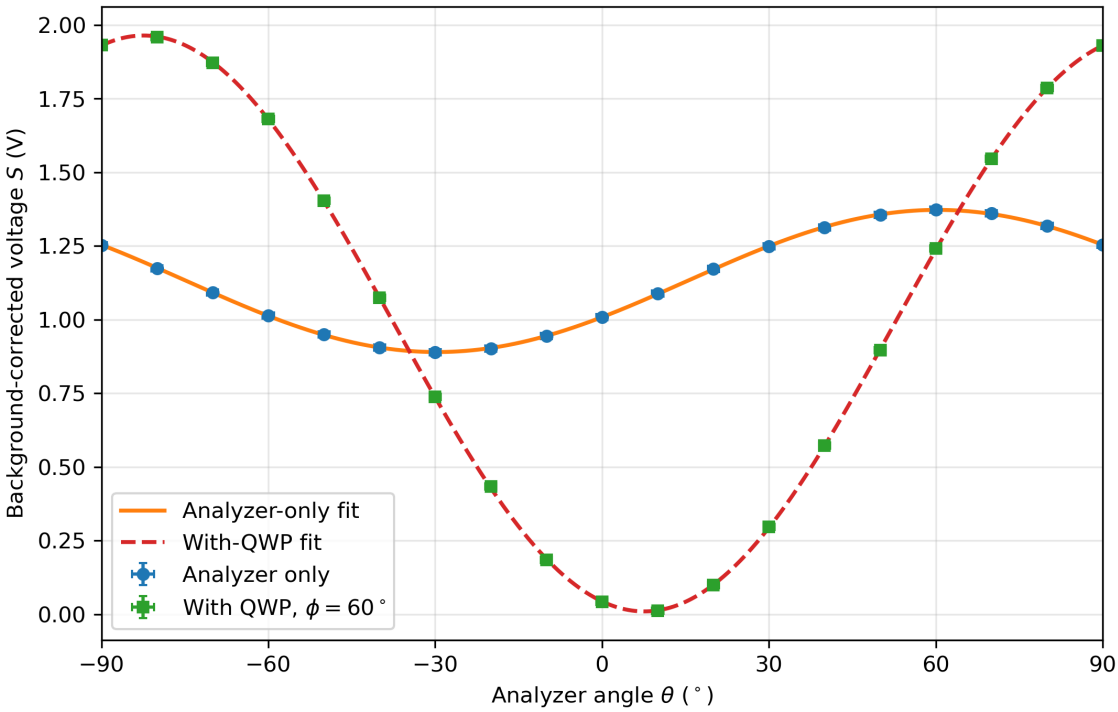


Figure 7: Background-corrected analyzer scans for BBB3 without and with the QWP at $\phi = 60°$.

5.7.3 Reasoning and Classification

The reasoning chain was

$$\begin{aligned} &\text{analyzer dependence with a large nonzero minimum} \\ &\quad \rightarrow \text{not purely linearly polarized} \\ &\quad \rightarrow \text{QWP insertion produces an almost zero minimum} \\ &\quad \rightarrow \text{the original fully polarized state was converted to linear} \\ &\quad \rightarrow \boxed{\text{elliptically polarized light}}. \end{aligned} \tag{149}$$

The large change in the fitted maximum angle,

$$60.3° \rightarrow -82.4°, \tag{150}$$

is not an inconsistency. A QWP changes the phase relation between the orthogonal components of an elliptical state and can therefore rotate the principal direction of the resulting linear polarization.

The handedness of the elliptical state was not determined because the measurement sequence was designed for classification rather than a complete measurement of the Stokes parameter s_3.

5.8 Uncertainty Analysis

The stated angular uncertainties were

$$\sigma_\theta = 1°, \qquad \sigma_\phi = 2°. \tag{151}$$

The fit itself returned much smaller formal uncertainties in the maximum directions, between approximately 0.02° and 0.09°. These values describe only the statistical precision with which a smooth sinusoidal curve can be fitted. They do not include scale reading, mechanical positioning, QWP alignment, or zero-angle errors and were therefore not used as the total experimental uncertainties.

The QWP orientation was selected from analyzer readings separated by 10°. Although the fitted maxima interpolate between the recorded points, the physical plate settings were based on the nearest measured maximum. This explains, for example, why BBB1 was set to $\phi = 80°$ although the fitted maximum was 84.1°.

Because repeated voltage measurements were not recorded, the point-to-point voltage uncertainty was estimated from the residual root-mean-square scatter:

$$\sigma_{U,\mathrm{emp}} = \sqrt{\frac{\sum_{i=1}^{N} \left[S_i - S_{\mathrm{fit}}(\theta_i)\right]^2}{N - p}}. \tag{152}$$

The empirical values ranged from approximately

$$0.0010\,\mathrm{V} \le \sigma_{U,\mathrm{emp}} \le 0.0045\,\mathrm{V}. \tag{153}$$

The largest relative uncertainty occurred near the almost extinguished minima of BBB2 and BBB3. For BBB2 without the QWP,

$$S_{\min} = 0.000687\,\mathrm{V} \tag{154}$$

was smaller than the empirical scatter

$$\sigma_{U,\mathrm{emp}} = 0.00196\,\mathrm{V}. \tag{155}$$

The precise numerical value of this minimum is therefore not significant; it is consistent with zero.
For BBB3 with the QWP,

$$\frac{\sigma_{U,\mathrm{emp}}}{S_{\min}} \approx \frac{0.00239}{0.00969} \approx 0.25. \tag{156}$$

The relative uncertainty of the minimum was therefore approximately 25%, but the minimum remained less than 0.5% of the maximum signal. The classification as nearly linear after the QWP is consequently robust.

For BBB1, both minima were large compared with the empirical voltage scatter. The conclusion that complete extinction was impossible was therefore not sensitive to detector noise.

The principal systematic error sources were:

- uncertainty in the analyzer and QWP angles,
- use of 10° measurement increments to select the QWP setting,
- imperfect polarizer extinction,
- QWP retardance differing from exactly $\pi/2$,
- approximate and wavelength-dependent QWP transmission,
- slow LED-output drift,
- beam displacement or clipping when the QWP was inserted,
- and nonuniformity of the unknown optical foils.

5.9 Overall Discussion

All six datasets covered a complete 180° analyzer interval and were accurately described by the expected sinusoidal dependence. The fits gave

$$R^2 > 0.9998 \tag{157}$$

for every scan, demonstrating that the measurements were internally consistent.

The estimated QWP losses were

$$10.2\%, \qquad 9.6\%, \qquad 12.7\% \tag{158}$$

for BBB1, BBB2, and BBB3, respectively. These values are physically reasonable and broadly consistent with the supervisor's approximate 10% estimate.

The classifications were supported by complete analyzer scans rather than isolated points. BBB2 was the most direct case because nearly complete extinction was already observed without the QWP. BBB3 was also classified unambiguously because the QWP converted a state with a large nonzero minimum into a nearly linearly polarized state. BBB1 was identified as a linear–unpolarized mixture because the QWP did not significantly alter its relative modulation or minimum ratio.

A complete determination of all Stokes parameters was not performed. Therefore, the measurements determine the polarization categories required by the operational method but do not provide the handedness of BBB3 or a full quantitative polarization ellipse.

5.10 Conclusion

The final classifications were

BBB1 :	mixture of linearly polarized and unpolarized light,
BBB2 :	linearly polarized light,
BBB3 :	elliptically polarized light.

(159)

The classifications were consistent with the operational procedure described in the experiment manual. The dominant relative uncertainty occurred near intensity minima, while the principal systematic uncertainties were the QWP alignment and the nonideal behavior of the optical elements.

6 Discussion

6.1 Agreement with the Theoretical Model

The measurements performed in Task 1 showed the expected transformation of initially linearly polarized light by a quarter-wave plate. For a plate orientation of

$$\phi = 0^\circ,$$

the measured analyzer dependence was almost completely modulated and closely followed the expected $\sin^2\theta$ behavior. The fitted modulation depth,

$$C = 0.9965,$$

was very close to the ideal value $C = 1$, showing that the light remained almost fully linearly polarized.

For

$$\phi = 30^\circ,$$

the measured modulation depth was

$$C = 0.4992,$$

in excellent agreement with the theoretical prediction

$$C_{\text{th}} = 0.5.$$

The measured maximum and minimum intensities were also close to the predicted values

$$S_{\max} = \frac{3S_0}{4}, \qquad S_{\min} = \frac{S_0}{4}.$$

The nonzero minimum confirmed that the outgoing light was elliptically polarized.

For

$$\phi = 45^\circ,$$

the ideal theory predicts circular polarization and therefore a signal independent of analyzer angle. The measured mean signal was close to one half of the reference intensity, as expected, but a small residual modulation remained:

$$C = 0.0501.$$

The output was therefore approximately circularly polarized but not perfectly circular. A deviation of the QWP angle by approximately 2° can itself produce a modulation of the same order, since

$$|\cos\left(2(45^\circ \pm 2^\circ)\right)| \approx 0.07.$$

The observed residual modulation is therefore compatible with the stated QWP alignment uncertainty. Additional contributions may have arisen from a retardance differing slightly from $\pi/2$, the spectral width of the LED, and imperfect input polarization.

All Task 1 curves were described very well by the sinusoidal fit model, with

$$R^2 > 0.999.$$

This confirms that the Jones-calculus description captures the principal angular dependence of the experiment.

6.2 Polarization Classification of the Black Boxes

The three unknown polarization states were classified using complete analyzer scans with and without the QWP. This method was sufficient for distinguishing the required polarization categories, although it did not provide all four Stokes parameters.

BBB1 showed a moderate analyzer dependence with a large nonzero minimum. Its modulation depth was almost unchanged after insertion of the QWP:

$$C_{\text{without}} = 0.1906, \qquad C_{\text{with}} = 0.1876.$$

The minimum-to-maximum ratio also remained nearly constant:

$$0.6798 \rightarrow 0.6841.$$

The QWP therefore reduced the overall signal but did not significantly change the polarization behavior. This is consistent with a mixture of linearly polarized and unpolarized light. In a simplified two-component model, the modulation depth suggests a linearly polarized fraction of approximately 19%, although this value should not be interpreted as a complete Stokes-based degree of polarization.

BBB2 showed almost complete extinction in the analyzer-only measurement:

$$\frac{S_{\min}}{S_{\max}} = 4.78 \times 10^{-4}.$$

The modulation depth was

$$C = 0.9990,$$

which is characteristic of nearly fully linearly polarized light. When the QWP was inserted approximately parallel to the measured polarization direction, the strong modulation remained. This is consistent with the expected behavior of linear polarization passing along one of the principal axes of a QWP.

BBB3 showed a moderate analyzer dependence with a large nonzero minimum before the QWP was inserted. This excluded purely linear polarization but did not initially distinguish elliptical polarization from a partially polarized mixture. After insertion of the QWP, the modulation depth increased from

$$0.2132$$

to

$$0.9902,$$

and the relative minimum decreased to

$$\frac{S_{\min}}{S_{\max}} = 0.00493.$$

The light could therefore be almost completely extinguished after the QWP. This demonstrates that the original state was fully elliptically polarized and was transformed into an approximately linear state.

The black-box classifications were therefore

$$\text{BBB1}: \text{ linear and unpolarized mixture},$$

$$\text{BBB2}: \text{ linear polarization},$$

and

$$\text{BBB3}: \text{ elliptical polarization}.$$

6.3 QWP Transmission and Signal Consistency

The supervisor estimated that the QWP reduced the transmitted intensity by approximately 10%. From the ratio of the fitted mean signals before and after insertion of the QWP, the measured losses were approximately

$$10.2\% \quad \text{for BBB1},$$

$$9.6\% \quad \text{for BBB2},$$

and

$$12.7\% \quad \text{for BBB3}.$$

The first two values agree very closely with the estimate. The somewhat larger loss measured for BBB3 is still reasonable because the 10% value was only approximate and the comparison may also include LED drift, changes in beam alignment, and position-dependent transmission through the optical elements.

The agreement between the two equivalent analyzer endpoints provided an additional sanity check. Since a linear analyzer has a 180° periodicity, measurements at the two ends of each full scan should be similar. The recorded curves satisfied this condition to within a small fraction of their total signal variation.

6.4 Uncertainty and Error Sources

The analyzer and QWP angular uncertainties were taken as

$$\sigma_\theta = 1^\circ, \qquad \sigma_\phi = 2^\circ.$$

Although the fits returned much smaller formal angular uncertainties, these fit uncertainties describe only the precision with which a sinusoidal curve can be interpolated. They do not include manual setting errors, parallax, zero-angle calibration, or mechanical reproducibility. The experimental angular uncertainties are therefore more appropriate for comparison with theory.

The fitted maximum positions in Task 1 differed from the theoretical values by

$$1.7^\circ$$

for $\phi = 0^\circ$ and

$$2.2^\circ$$

for $\phi = 30^\circ$. The combined angular uncertainty is approximately

$$\sqrt{\sigma_\theta^2 + \sigma_\phi^2} \approx 2.2^\circ,$$

so both deviations are compatible with the theoretical predictions.

Because repeated voltage measurements were not recorded, the random voltage uncertainty was estimated from the residual scatter around the fitted curves. The empirical residual RMS values were typically of the order of a few millivolts. This procedure gives a reasonable estimate of point-to-point fluctuations but does not replace a direct statistical uncertainty obtained from repeated measurements.

The largest relative uncertainty occurred near intensity minima. For example, the fitted minimum in the Task 1 $\phi = 0^\circ$ scan was only a few millivolts, comparable to the residual voltage scatter. The relative uncertainty near extinction was therefore very large, even though the absolute uncertainty was small. At high signal levels, the same voltage uncertainty corresponded to a relative uncertainty below one percent.

The most important systematic error sources were uncertainty in the analyzer and QWP angles, incomplete extinction by the real polarizers, a QWP retardance differing from exactly $\pi/2$, the finite spectral bandwidth of the 590 nm LED, slow drift of the LED intensity, detector background and stray light, slight beam displacement when optical elements were inserted or rotated and possible nonuniformity of the black-box optical foils.

For Task 1, the QWP alignment and nonideal retardance were the most important systematic effects when comparing the measured curves with ideal theory. For measurements near extinction, detector fluctuations and background subtraction had the largest effect on the relative uncertainty.

6.5 Limitations of the Measurement Method

The analyzer scans were recorded at 10° intervals. This was sufficient to identify the sinusoidal form and determine the polarization class, but it limited the direct experimental resolution of the extrema. The fitted maximum positions interpolate between measured points and should therefore not be interpreted as direct angle readings with sub-degree accuracy.

The Task 2 procedure classified the unknown states using the operational guide rather than a complete Stokes-parameter analysis. Consequently, the polarization categories were determined reliably, but some information remained unavailable. In particular, the handedness of the elliptically polarized state produced by BBB3 could not be established, and the exact polarization ellipse was not reconstructed.

The classification of BBB1 is also less direct than those of BBB2 and BBB3. The nearly unchanged modulation after insertion of the QWP strongly supports a linear–unpolarized mixture, but a complete Stokes analysis would provide a more quantitative determination of the polarized fraction.

6.6 Possible Improvements

The experimental reliability could be improved by recording repeated voltage measurements at every analyzer angle. This would allow the random uncertainty to be calculated directly from the standard deviation rather than estimated from fit residuals.

Smaller angular steps near maxima and minima would improve the determination of the principal polarization directions. A motorized rotation stage would further reduce positioning uncertainty and mechanical backlash.

The background voltage could be measured before and after each scan to detect slow detector or ambient-light drift. A second reference photodiode could also be used to monitor variations in the LED output.

A narrower-band or monochromatic light source would reduce the effect of the wavelength dependence of the QWP retardance. Finally, measuring the full set of Stokes parameters would permit a quantitative determination of the degree of polarization, the polarization ellipse, and the handedness of the elliptical state.

6.7 Overall Assessment

The experiment successfully demonstrated the operation of a linear analyzer and a quarter-wave plate. The Task 1 measurements agreed closely with the theoretical predictions for linear and elliptical polarization, while the small deviation from ideal circular polarization was consistent with the experimental angular uncertainty and nonideal optical components.

The black-box measurements produced clear and physically plausible classifications. The strong agreement of the fitted curves with the sinusoidal model, the consistent QWP transmission losses, and the successful extinction tests for BBB2 and BBB3 show that the measurement procedure was reliable.

Within the stated uncertainties and methodological limitations, the experimental results were consistent with the Jones-calculus description of polarized light and with the operational polarization-classification method.

7 Conclusion

The polarization of light was investigated using a rotatable linear analyzer and a quarter-wave plate. In Task 1, initially linearly polarized light was measured after passing through the quarter-wave plate at

$$\phi = 0^\circ, \qquad 30^\circ, \qquad 45^\circ.$$

The fitted modulation depths were

$$C = 0.9965, \qquad 0.4992, \qquad 0.0501,$$

respectively. These values agreed closely with the theoretical predictions of 1, 0.5, and 0. The corresponding output states were therefore identified as nearly linear, elliptical, and approximately circular polarization. The small residual modulation at $\phi = 45^\circ$ was consistent with the QWP angular uncertainty and nonideal optical behavior.

In Task 2, the polarization states produced by three unknown optical elements were characterized using complete analyzer scans with and without the quarter-wave plate. BBB1 was identified as a mixture of linearly polarized and unpolarized light because the relative modulation remained almost unchanged after insertion of the QWP. BBB2 produced linearly polarized light, as nearly complete extinction was observed with the analyzer alone. BBB3 produced elliptically polarized light because the QWP transformed the initial state into an almost completely extinguishable, approximately linear state.

The measurements were well described by the expected sinusoidal analyzer dependence, with all fitted curves yielding

$$R^2 > 0.999.$$

The largest relative uncertainty occurred near intensity minima, where the remaining signal was comparable to the detector fluctuations and background correction. The principal systematic uncertainties arose from the analyzer and QWP alignment, imperfect optical extinction, and deviations of the QWP retardance from its ideal value.

Overall, the experimental results were consistent with the theoretical description of polarized light using Jones calculus and demonstrated the use of a linear analyzer and quarter-wave plate for distinguishing linear, elliptical, partially polarized, and approximately circular polarization states.

8 References

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